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| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 02/01/2016 |
| **PI Name:** | Eisch, Amelia Ph.D. | | |
| **Project Title:** | Molecular and Cellular Mechanisms of Space Radiation-Induced Changes in Hippocampal-Dependent Behavior | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 19104 | **Congressional District:** | 3 |
| **Comments:** | NOTE: Previously at University of Texas Southwestern Medical Center at Dallas, 2000-2016 | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2011 Space Radiobiology NNJ11ZSA001N |
| **Start Date:** | 11/10/2011 | **End Date:** | 11/09/2015 |
| **No. of Post Docs:** | 8 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 6 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 6 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
| **Contact Email:** | lisa.c.simonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: New end date of 11/9/2015 per NSSC information (Ed., 8/19/14) | | |
| **Key Personnel Changes/Previous PI:** | None. | | |
| **COI Name (Institution):** | Chen, Benjamin ( University Of Texas, Dallas ) | | |
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| **Performance Goal Text:** | | | |

An unavoidable consequence of many space missions is exposure to galactic cosmic radiation (GCR), which includes HZE particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) is a high research priority. Here we propose extensive analysis of how a specific region of the CNS, the hippocampus, is influenced by radiation. The hippocampus is important because it contributes to spatial learning and mood regulation, which are critical for mission success. Recently the hippocampal dentate gyrus (DG) was shown to be involved in learning challenging tasks requiring fine spatial distinction ("pattern separation"). The hippocampus is also important because it is exceptionally "neuroplastic." For example, in terms of gene transcription changes, the DG is among the most responsive CNS regions to physiological and environmental stimuli. The DG is also a site of neurogenesis throughout life, and adult neurogenesis is proposed to contribute to hippocampal function in general and pattern separation in particular.

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| **Task Description:** | Understanding how space radiation influences DG structure and function will allow far better estimation of CNS radiation risks. Research already suggests that space radiation has a negative effect on hippocampal structure and function. However, it is unknown whether space radiation specifically influences DG functions like pattern separation. It is also unknown whether the diverse neural cells types within the DG are similarly influenced by space radiation or whether, for example, there is a cell-type specific change in translation or morphology. We will use recent advances in knowledge of DG function and structure and cutting-edge but proven technical approaches to answer key remaining questions about how space radiation influences the cellular, molecular, and functional integrity of the hippocampus.<br>For example in Aim 1, using behavioral tests with which we have recently published (delayed non-match to pattern radial arm maze for fine spatial distribution; Guo et al., Nat Med 2011), we will explore our hypothesis that space radiation decreases pattern separation in a dose- and LET-dependent manner.<br><br>In Aim 2, using transgenic mice we have developed that allow inducible YFP labeling of neural stem cells and their progeny (nestin-CreERT2/R26R mice; Lagace et al., J Neurosci 2007), we will explore our hypothesis that space radiation negatively impacts early postnatally-generated DG neurons.<br><br>In Aim 3, using transgenic mice developed by our colleagues that allow profiling of translating RNA in genetically-defined DG neural cells, we will explore our hypothesis that space radiation causes discrete changes in DG neural cell types in a dose- and LET-dependent manner. Unlike currently used approaches to assess total RNA profiles from cellularly heterogeneous samples (e.g., laser capture dissection and subsequent microarray analysis), this approach allows purification of specific cell types via ribosome tag and a quantitative comparison of translating RNA profiles in genetically-distinct DG cells.<br><br>Importantly, all mice will be 6-7 months old at time of radiation exposure in order to be approximately the age equivalent to a 30 year old human, and thus to better assess the impact of ground-based space radiation on the fully mature brain. Tissue from Aims 1-3 will be examined using immunohistochemistry, epifluorescent and confocal microscopy, stereologic cell counting, and morphological analysis, and tissue from Aim 3 will be explored using microarrays and RNAseq on the translating RNA pools and qRT-PCR.<br><br>Taken together, these aims will allow us to test our overall hypothesis that radiation will have distinct, long-lasting effects on the behavioral, cellular, and molecular integrity of the hippocampus. Thus, these results will expedite the quest to estimate CNS radiation risks due to GCR exposure. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | It is critical to identify factors that regulate normal brain function and structure. Our proposed research will elucidate how ground-based space radiation influences hippocampal structure and function. The knowledge derived from the proposed experiments will provide insight into the mechanisms contributing to brain and hippocampal function and their disruption by space radiation, and thus has the potential to shed light on how we can estimate CNS risk from space radiation. This work is important given the long-duration space missions on the former Mir Space Station, the present International Space Station, and the potential exploratory missions to the Moon and Mars. Mice will be used as the primary source of nervous tissue for experimentation since most of the relevant literature concerns studies of tissues of rodent origin, and since the mouse is an excellent model for human exposure to space radiation given their remarkably similar brain structure and function and genomic similarity. At this time, using computer modeling to understand the effects of space radiation on behavioral, cellular, and molecular indices of the hippocampus is just not possible given the complexity of the CNS. Cell culture systems have proven extremely useful to ask mechanistic questions about the effects of space radiation on the hippocampus, but they are unable to address the effects of space radiation on the whole organism, particularly behavior. The combination of approaches used here -- analysis of behavior, cellular morphology, and translational profile -- will have significant benefits for the fields of neuroscience in general and the fields of learning and memory, stem cell biology, and even neural development in particular, but are all inspired by the need to gain greater insight into whether the CNS alterations experienced by astronauts may compromise mission success.<br>The particular focus of this application -- hippocampal dysfunction -- is evident in a range of psychiatric and neurologic disorders, including Alzheimer's Disease, depression, and addiction. Astronauts may be particularly at risk for delayed hippocampal pathological signs, as it is estimated that even just 1 year in a low-Earth orbit mission may result in 91% of hippocampal cells and 25% of hippocampal cell-nuclei hit by high LET particles. These studies will help us predict neurodegenerative or cognitive changes in astronauts during the mission (which would compromise mission success) or even after mission completion. We also focus on the hippocampus since it is exceptionally "neuroplastic" – being among the most responsive brain regions in terms of gene transcription after physiological manipulation and stimuli and being a site of the birth of new neurons throughout life. This exceptional neuroplasticity suggests that the hippocampus may be among the first brain regions to reveal a response to radiation. To best study the influence of ground-based space radiation on hippocampal structure and function in a way that is most translational from rodents to humans, we will be specifically using older mice (e.g., ~6-7 months old at time of irradiation, which is equivalent to 30 human years).<br><br>Aside from the general benefits of this broader work for neuroscience and humans, there are several innovative aspects to this work. For example, using mice that we have generated that allow inducible labeling and tracking of embryonic vs. adult-generated neurons, we will be able to for the first time tease apart effects on these two populations. In addition, using novel mice that allow profiling of translating RNA in genetically-defined DG neural cells (e.g., adult-generated vs. embryonic-generated DG neurons), we will make a quantitative comparison of translating RNA profiles in genetically-distinct DG neural cells in a time-, dose-, and LET-dependent manner. Utilization of these novel technologies have the implication to expand our broader understanding of mechanisms underlying neuroplasticity and even neurodegenerative processes. |

Aim 1: Determine the impact of low- and high-LET radiation on hippocampal function.
Given past published work in the literature, we hypothesized that space radiation would diminish hippocampal function. However we noticed that most published studies irradiated animals when they are very young or only young adults, and that no studies examined very challenging behaviors such as context discrimination. Therefore for this aim, we irradiated animals when they were young adults or when they were mature, and tested them on both relatively simple tasks, such as fear conditioning, and more challenging tasks such as contextual discrimination. Our data show that context discrimination fear conditioning (CDFC) is differentially influenced in mice that are young adult vs. mature at the time of irradiation. Specifically, mice that are mature at the time of irradiation learn more consistently and faster to discriminate two contexts, while mice that are younger at the time of irradiation generally show decreased context separation. We find this is true for both 28Si and 56Fe, with some dose- and delivery-dependency.

Our data also find that contextual fear conditioning (CFC) is differentially influenced by the age of the mice at the time of irradiation. For example, in general mice that are young adult at the time of irradiation show diminished CFC, while mice that are mature at the time of irradiation show no difference. We find this is true for both 28Si and 56Fe, although again there is some dose- and delivery-dependency.

Our data show no gross difference in other behavioral and physiological measures, including weight, locomotor activity, exploration of a novel environment or dark vs. light environments, and pain threshold.

Given that irradiation induced improvement in contextual discrimination, and given that improved contextual discrimination is seen after induced increase in the number of new neurons born in hippocampus, we next hypothesized that neurogenesis would be enhanced in the mice that performed better. In keeping with published data with other types of space radiation, our data show that mice that received 28Si irradiation in young adulthood had decreased neurogenesis. However mice that received fractionated 56Fe irradiation in young adulthood had no change in neurogenesis at the time point examined. Interestingly, and in contrast with our hypothesis, my step received the radiation when mature showed either decreased neurogenesis (silicon) or no change in neurogenesis (iron). This disproves our initial hypothesis that space radiation-induced improvement in CDFC was reflective of increased hippocampal neurogenesis (or less of a decrease in hippocampal neurogenesis relative to controls) in mice that were mature at the time of irradiation.

Aim 2: Determine how radiation influences the dendritic morphology of embryonic- vs. adult- generated granule cell neurons.

We have collected multiple cohorts of Nestin-CreERT2/R26R-YFP mice that received Tamoxifen at postnatal day 3 and were irradiated in adulthood. These tissues have been examined for the effect of both 28Si and 56Fe on dendritic morphology in cells that were generated in the early postnatal period. We will complete these analyses prior to unblinding the coded tissue. We also have tissue from animals exposed to O and C that will be examined for related measures so we can publish a comparative study for this aim.

Aim 3: Define the impact of radiation on the translational profile of genetically defined DG neural cells.

We have collected multiple cohorts of BAC-TRAP mice that were irradiated in adulthood. These tissues are being currently examined for the effect of both 28Si and 56Fe on RNAseq and microarray expression. We have recently received the results from the RNA libraries we submitted and are performing the analysis now.

Publications

We made excellent progress on publications, as we already have 15 articles published, and at least 9 more in preparation.

Publications linked to these NASA grants:

1. Decarolis NA, Mechanic M, Petrik D, Carlton AF, Malhotra S, Ables JL, Bachoo R, Lagace DC, Eisch AJ. In vivo contribution of nestin- and GLAST-lineage cells to adult hippocampal neurogenesis. Hippocampus. 23: 708–719. doi: 10.1002/hipo.22130 (2013). PMID 23554226, PMC 3718551.

2. Eisch AJ, Petrik D. Neurogenesis and depression: a road to remission? Science. (2012) Oct 5;338(6103):72-5. doi: 10.1126/science.1222941. PMID 23042885, PMC 3756889.

3. Petrik D, Lagace DC, Eisch AJ. The neurogenesis hypothesis of affective and anxiety disorders: are we mistaking the scaffolding for the building? Neuropharmacology. Jan;62(1):21-34. Epub 2011 Sep 19 (2012) PMID 21945290, PMC 3698048.

4. Latchney SE and Eisch AJ. Therapeutic application of neural stem cells and adult neurogenesis for neurodegenerative disorders: regeneration and beyond. European Journal of Neurodegenerative Diseases. 1(3): 335-351 (2012). PMID 25729743, PMC 4340249.

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7. Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Lucero MJ, DeCarolis NA, Chen BPC, Eisch AJ. Acute and Fractionated Exposure to High-LET 56Fe HZE Particle Radiation Both Result in Similar Long-Term Deficits in Adult Hippocampal Neurogenesis. Radiation Research. (Epub Dec 9 2013) Dec;180(6):658-67. doi: 10.1667/RR13480.1. (2013) PMID 24320054, PMC 3930069.*authors contributed equally to this work.

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9. Latchney SE, Rivera PD, Mao XW, Ferguson VL, Bateman TA, Stodieck LS, Nelson GA, Eisch AJ. The effect of

**Task Progress:**

spaceflight on mouse olfactory bulb volume, neurogenesis, and cell death indicates the protective effect of novel environment. Journal of Applied Physiology. 2014 Apr 17. [Epub ahead of print] PMID 24744382, PMC Journal - In Process (Publisher Provided).

10. Walker AK, Rivera PD, Wang Q, Chuang JC, Tran S, Osborne-Lawrence S, Estill SJ, Starwalt R, Huntington P, Morlock L, Naidoo J, Williams NS, Ready JM, Eisch AJ, Pieper AA, Zigman JM. The P7C3 class of neuroprotective compounds exerts antidepressant efficacy in mice by increasing hippocampal neurogenesis. Molecular Psychiatry. 20(4): 500-508 (2014). PMID 24751964, PMC 4206684.

11. DeCarolis NA*, Rivera PD*, Ahn HF, Shih H-Y, Amaral WZ, LeBlanc JA, Malhotra S, Mukherjee S, Chen BPC, Eisch AJ. 56Fe particle exposure results in long-lasting genomic instability and transiently suppresses adult hippocampal neurogenesis in vivo. Life Sciences in Space Research. 2:70-79 (2014). PMID 25170435, PMC 4142527. *equal contribution, listed alphabetically

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13. Latchney SE*, Jiang Y*, Petrik D, Eisch AJ^, Hsieh J^. Inducible knockout of Mef2A/C/D from nestin-expressing stem/progentior cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo. The Journal of the Federation of American Societies for Experimental Biology (The FASEB Journal). 2015 Aug 18. pii: fj.15-275651 13pp. PMID 26286136, PMC in progress. *equal contribution, ^co-corresponding authors

14. Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells. Stem Cells. 2015 Sep 29. doi: 10.1002/stem.2215. 11pp. PMID 26418130, PMC in progress. *equal contribution, listed alphabetically, ^co-corresponding authors

15. Yun S, Donovan MH, Noonan MA, Richardson DR, Reister R, Farnbauch LF, Fisher SJ, Riethmacher D, Gershenfeld HK, Lagace DC^, Eisch AJ^. Stress induced Anxiety- and Depressive-like Phenotype Associated with Transient Reduction in Adult Nestin-CreERT2/Diphtheria Toxin Fragment A Transgenic Mice. Accepted, PLoS One. ^co-corresponding authors Articles in revision or preparation:

1. Rivera PD, Reynolds RP, Just AL, Birnbaum SG, Eisch AJ. Image-guided cranial irradiation-induced ablation of dentate gyrus neurogenesis diminishes extinction of young – but not old – morphine reward memories. In revision for resubmission to Hippocampus.

2. Bulin SE, Richardson DR, Mendoza ML, Zhang Z, Song KH, Solberg TD, Eisch AJ. Ablation of dentate gyrus neurogenesis via cranial irradiation enhances morphine self-administration and morphine-induced locomotor sensitization. In revision for resubmission to Biological Psychiatry.

3. Walker AK*, DeCarolis NA*, Ahn HF, Lagace DC, Ables JL, Redfield RL, Cole MG, Chen BPC, Eisch AJ. Running after irradiation ameliorates the proliferation deficit but not the nestin-expressing Type-1 cell deficit in the adult hippocampal subgranular zone. In revision for resubmission to Hippocampus. *equal contribution

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5. Walker AK*, Whoolery CW*, Richardson DR, Reynolds RP, Rivera PD, Shih H-Y, Redfield RL, Lucero MJ, Beddow DH, Mukherjee S, LeBlanc JA, Cole MG, Amaral WZ, Malhotra S, DeCarolis NA, Chen BPC, Eisch AJ. Whole body exposure to high LET 28Si particles decreases hippocampal progenitor proliferation and neurogenesis in the short-term, and new neuron survival in the long-term. In preparation for submission to Radiation Research. *equal contribution, listed alphabetically.

6. Lucero MJ,* Walker AK,* Whoolery CW,* Redfield RL, Ito N, Richardson DR, Reynolds RP, Palchik G, Mukherjee S, Shih H-Y, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Space radiation unexpectedly enhances pattern separation yet diminishes dentate gyrus neurogenesis in a dose- and age-dependent manner. In preparation. *equal contribution, listed alphabetically

7. Yun S, Lucero MJ, Ito N, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Molecular basis for space radiation-induced deficits hippocampal function in aged animals. In progress

8. Cole MG, Shih H-Y, LeBlanc JA, Rivera PD, Chen BPC, Eisch AJ. Short and long term effects of 16O radiation on adult hippocampal neurogenesis and indices of genomic instability. In progress

9. Cole MG, Shih H-Y, LeBlanc JA, Chen BPC, Eisch AJ. A model of acute space radiation exposure (12C) and adult hippocampal neurogenesis. In progress

We have also presented three oral/poster presentations dedicated to the findings of this grant in the final year of the grant:

1. Lucero MJ, Redfield RL, Ito N, Reynolds RP, Richardson DR, Palchik G, Mukherjee S, Shih HS, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Neural Mechanisms Contributing to HZE Radiation-Induced Improvement in Pattern Separation in Adult Mice NASA 2015 Human Research Program Investigators' Workshop, Galveston TX, February 2015.

2. Walker AK*, Whoolery CW*, Richardson DR, Reynolds RP, Rivera PD, Shih H-Y, Redfield RL, Lucero MJ, Beddow DH, Mukherjee S, Chen BPC, Eisch AJ. Galactic cosmic radiation (28Si) reduces dentate gyrus neurogenesis in the long-term in a dose-dependent manner. The Annual Meeting of the Society for Neuroscience, Chicago, IL October 2015. *equal contribution, listed alphabetically.

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| | 3. Eisch AJ, Lucero MJ, Redfield RL, Ito N, Richardson DR, Reynolds RP, Palchik G, Mukherjee S, Walker AK, Whoolery CW, Shih H-Y, Rivera PD, Birnbaum SG, Chen BPC. What happens to mouse hippocampal-dependent behavior and neurogenesis on the way to Mars? One small step for mousekind. The Annual Meeting of the Society for Neuroscience, Chicago, IL October 2015. |
| **Bibliography Type:** | Description: (Last Updated: 11/17/2025) |
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| **Articles in Peer-reviewed Journals** | Latchney SE*, Jiang Y*, Petrik D, Eisch AJ^, Hsieh J^. (*equal contribution, ^co-corresponding authors) "nducible knockout of Mef2a, -c, and -d from nestin-expressing stem/progenitor cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo." FASEB J. 2015 Dec;29(12):5059-71. Epub 2015 Aug 18. http://dx.doi.org/10.1096/fj.15-275651 ; PubMed PMID: 26286136; PubMed Central PMCID: PMC4653059 , Dec-2015 |
| **Articles in Peer-reviewed Journals** | Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. (*equal contribution, listed alphabetically, ^co-corresponding authors) "Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells." Stem Cells. 2015 Dec;33(12):3655-65. Epub 2015 Oct 9. http://dx.doi.org/10.1002/stem.2215 ; PubMed PMID: 26418130; PubMed Central PMCID: PMC4713255 , Dec-2015 |
| **Articles in Peer-reviewed Journals** | Yun S, Donovan MH, Richardson DR, Reister R, Farnbauch LA, Fischer SJ, Riethmacher D, Gershenfeld HK, Lagace DC^, Eisch AJ^. (^co-corresponding authors) "Stress-induced anxiety- and depressive-like phenotype associated with transient reduction in neurogenesis in adult Nestin-CreERT2/diphtheria toxin fragment A transgenic mice." PLoS One. 2016 Jan 21;11(1):e0147256. eCollection 2016. http://dx.doi.org/10.1371/journal.pone.0147256 ; PubMed PMID: 26795203; PubMed Central PMCID: PMC4721672 , Jan-2016 |
| **Articles in Peer-reviewed Journals** | Whoolery CW, Walker AK, Richardson DR, Lucero MJ, Reynolds RP, Beddow DH, Clark KL, Shih HY, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Ahn F, Bulin SE, DeCarolis NA, Rivera PD, Chen BPC, Yun S, Eisch AJ. "Whole-body exposure to 28Si-radiation dose-dependently disrupts dentate gyrus neurogenesis and proliferation in the short term and new neuron survival and contextual fear conditioning in the long term." Radiat Res. 2017 Nov;188(5):532-551. Epub 2017 Sep 25. https://doi.org/10.1667/RR14797.1 ; PubMed PMID: 28945526 , Nov-2017 |
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